Introduction

This page describes the **modeling guidelines for IT Components** in the context of enterprise architecture at Porsche. It is based on the general modeling guidelines of PAG and includes specific instructions for creating and maintaining IT Component in LeanIX. The creation and editing of an IT Component fact sheet follows the standard procedure for creating a general fact sheet as described in "How to create a fact sheet".

IT components are modeled to describe the technical architecture of an IT product. In addition to infrastructure, platform, and service components required by the IT product, there must be at least one IT component that represents the technical characteristics of the product—such as its version, vendor lifecycle, or manufacturer.

IT components can only be created by an authorized group of users. Currently, this group consists of the administrators and the users with the role "Architect"

IT-Component or IT Product?

One of the most common challenges in architecture modeling is determining whether a given object should be classified as a **technical IT component** or a **business-facing IT product**.

Key Characteristics of an IT Component

An IT component typically meets the following criteria:

- It is a hardware or software unit with a clearly defined and dedicated use.
- It can be installed, enabled, disabled, or removed from an operational perspective.
- It represents the smallest modelable unit that forms part of an information system.
- It can be independently modified, distributed, or replaced by third parties.
- It has a managed vendor or manufacturer support lifecycle.
- It has a known vendor, manufacturer, or developer.
- It may incur costs (e.g., purchase, licensing, support).
- It may be subject to service-level agreements or operational targets.

Checklist for Classifying Architecture Objects

The following criteria help distinguish whether a given object should be classified as a **technical IT component** or a **business-facing IT product**. Use the checklist below to decide whether an object should be modeled as an **IT Component**, an **IT Product**, or is **not relevant for architecture modeling**.

🧭 Decision Questions

Question	If YES →	If NO →
1. Does the object provide purely technical functionality without business logic?	→ IT Component	→ Proceed to Question 2
2. Does the object directly support a business process?	→ IT Product	→ Proceed to Question 3
3. Is the object organizational, physical, or contractual in nature?	→ Not an IT Component	→ Proceed to Question 4
4. Is the object actively used by end users?	→ Likely an IT Product	→ Possibly technical infrastructure
5. Does the object consist of multiple technical components?	→ IT Product or IT Service	→ Evaluate as a single component
6. Can the object be documented in architecture tools like LeanIX?	→ Check if it fits as Product or Component	→ Likely not directly modelable

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📌 Example Classifications

The following examples illustrate how to distinguish between IT Components, IT Products, and non-architecture-relevant objects. These classifications help ensure consistent documentation and modeling in enterprise architecture tools such as LeanIX. Use them as guidance when evaluating new or existing elements in your IT landscape.

Object	Classification	Reason
Microsoft SQL Server	IT Component	Technical platform without business logic
SAP S/4HANA	IT Product	Supports business processes like finance and logistics
Customer (data object)	Not an IT Component	Represents business data, not technical functionality
Azure Active Directory	IT Component	Technical security component
Jira	IT Product	Used by end users to support business workflows
Project “CRM Migration”	Not an IT Component	Temporary organizational construct

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Structure Summary

Field	Content
Section Title	Interface Fact Sheet Modeling Guideline
Purpose	To guide users in distinguishing between IT Components and IT Products in LeanIX modeling.
Target Roles	Enterprise Architects, Business Analysts, Business Stakeholders
Key Concepts	IT Component, IT Product, Technical Architecture, Vendor Lifecycle, Modeling Criteria
IT Component Definition	A technical unit (hardware or software) with a defined lifecycle, vendor, and operational independence.
Modeling Requirement	Every IT Product must be linked to at least one IT Component that reflects its technical characteristics.
Decision Support	Includes a structured checklist and real-world examples
Common Challenge	Differentiating between business-facing and technical elements
Tool Context	LeanIX – Interface Fact Sheet
Language	English (optimized for clarity, consistency, and LLM training)
Use Cases	Architecture documentation, modeling governance, onboarding, training
LLM Readiness	Structured, role-specific, terminology-aligned, and example-driven content suitable for training and automation

General Modeling Rules

Subscriptions

In LeanIX, **subscriptions** are used to assign individuals and roles to a Fact Sheet. There are two main types of subscriptions:

- **Responsible**: Individuals accountable for the content and maintenance of the Fact Sheet.
- **Observer**: Individuals who are informed but not responsible for the content.

The specific roles that can be assigned to a Fact Sheet are defined in the modeling guidelines for each Fact Sheet type.

Responsibility Requirement

Each Fact Sheet must have **at least one person assigned as Responsible**. This person (or persons):

- Is accountable for the **accuracy, completeness, and currency** of the Fact Sheet.
- Receives **notifications** about changes to the Fact Sheet.
- Acts as the **primary contact** for questions, clarifications, or requirements related to the Fact Sheet.

Change of Responsibility

If a Responsible person:

- Changes roles,
- Leaves the company or department,
- Or is no longer involved with the Fact Sheet,

…it is their duty to ensure a **successor** is assigned to maintain the Fact Sheet.

Automated Quality Checks

LeanIX includes automated processes to ensure data quality. These processes include:

1. **Inactive Responsible Person**
   If a Responsible person is no longer active in the company or subsidiary (e.g., due to departure), the system checks for other Responsible persons:
   - If none exist, the **direct manager** of the former Responsible is identified and automatically assigned as the new Responsible.
   - The manager also receives a **To-Do** to assign a new, appropriate Responsible person.
   - This escalation continues up to the **department head level** if necessary.
2. **No Responsible Assigned**
   If no Responsible is assigned to a Fact Sheet:
   - The system uses the **"Responsible IT Department"** relation to identify the appropriate manager.
   - This manager is then assigned as the Responsible person and is expected to **nominate a qualified business owner**.

Lifecycle Management in LeanIX Fact Sheets

Each **Fact Sheet type** in LeanIX can be assigned a **lifecycle**. Whether this assignment is **mandatory** depends on the specific Fact Sheet type and is indicated on its detail page.

Recommended Practice

Regardless of whether the lifecycle is mandatory, it is **strongly recommended** to provide the following information:

- **Start of productive use** (Go-Live date)
- **Decommissioning date** (if known)

Depending on the desired level of detail, additional lifecycle milestones can also be modeled (e.g., development start, pilot phase, sunset planning).

Benefits

Accurate lifecycle data supports:

- **Strategic planning** and **portfolio management**
- **Compliance** with regulatory and audit requirements
- **Impact analysis** for changes and decommissioning
- **Architecture governance** across all layers

Organizational Responsibilities

Each Fact Sheet in LeanIX must include an assignment of **organizational responsibility**. This responsibility can be defined at two levels:

- **Organizational Structure Level** (e.g., departments, business units)
- **Process Organization Level** (e.g., Agile Release Trains, portfolios)

For certain Fact Sheet types, a distinction may also be made between **functional (business) responsibility** and **technical (IT) responsibility**.

> ✅ At least one form of responsibility must be specified—either at the organizational structure level or the process organization level.
> 🔄 Ideally, both levels should be documented to ensure full traceability and accountability.
>
> ⚠️ **If IT responsibility is defined as mandatory in the modeling guidelines for a specific Fact Sheet type, it must be provided without exception.**

Responsibility at the Organizational Structure Level

At Porsche, the **organizational structure** is defined by the official organizational chart, which organizes the company into hierarchical levels such as:

- Division
- Main Department
- Department
- Subdepartment (lowest level)

> In LeanIX modeling, unless explicitly stated otherwise, responsibility must always be assigned to the **lowest available level** of the organizational structure.

Key Principles

- Always reference the **most granular organizational unit** available.
- Ensure that the assigned unit reflects the **actual operational responsibility**.
- This ensures **clear accountability**, **accurate reporting**, and **effective governance**.

Responsibility at the Process Organization Level

Many IT products and applications at Porsche are developed and operated by **Agile Release Trains (ARTs)**. Others are managed within specific **portfolios**.

Assignment Guidelines

- If an **ART** is responsible for a product (or any other relevant Fact Sheet type), this responsibility must be **explicitly documented** in LeanIX by assigning the corresponding ART Fact Sheet.
- If the responsibility lies with a **single portfolio**—as is often the case with **P51 applications**—then assigning the portfolio is **sufficient**.

> ✅ Always ensure that the assigned ART or portfolio accurately reflects the **operational ownership** of the product.

Key Principles

- Use the **ART Fact Sheet** when agile teams are responsible for development and operations.
- Use the **Portfolio Fact Sheet** when responsibility is anchored in a business or IT portfolio.
- This ensures **clear accountability**, supports **agile governance**, and aligns with **LeanIX modeling standards**.

Please note

Only organizational fact sheets that are tagged with "**Synced**" or "**Manual**" may be used! The use of fact sheets with the tag "**Legacy**" is not allowed!

Modeling Version History with Successor and Predecessor Relations

In LeanIX, the **Successor** and **Predecessor** relations are used to model the **version history** of Fact Sheets over time.

These relations should be applied whenever a **new version** of an existing Fact Sheet is created to **replace** an older version.

> ✅ Successor relations can also be used **while the previous version is still active** and in productive use.

Best Practice for Unused Relations

If no predecessor or successor exists for a given Fact Sheet, it is recommended to **explicitly set the relation field to**:

> **"Leave empty"**

This ensures clarity in the data model and avoids ambiguity in version tracking.

Key Principles

- Use **Successor/Predecessor** to document version transitions.
- Apply even if the older version is still operational.
- If not applicable, set the relation field to **"Leave empty"** instead of leaving it undefined.

Specific Modelling Rules and Best Practices for an IT Component

Authorization to create a component

Only authorized users (currently: administrators and users with the "Architect" role) are permitted to create IT components in LeanIX.
If a product manager identifies a missing component or a new version is required, a request must be submitted via the LeanIX Help Center. All required attributes must be provided accurately.

Responsibility for IT Components

The definition of who is responsible for an IT component continues to lead to heated discussions in the company. Especially due to the establishment of the so-called departmental IT (P51), we are in a dilemma here from an IT point of view. The person specified on a component is typically the person responsible for that component for the entire organization in terms of operations, maintenance, security, and compliance. This regulation is also not a problem for centrally provided components, but it causes problems in many other contexts and poses challenges for IT. We have not yet found a suitable solution to the problem, but we are working on a solution.

The person assigned as "Responsible" for an IT component is accountable for:

- Operation
- Maintenance
- Security
- Compliance

This applies company-wide. Creating or requesting a component without assigning a responsible person is not permitted and will be rejected.

Creating a new IT component

When a product, tool, or software is identified as a technical IT component, it must be modeled using an **IT Component Fact Sheet**. Required attributes and relations:

- **Category**
- **Version (major only)**
- **Description (in English)**
- **Lifecycle**
- **Vendor assignment**

IT Component Categories

Each component must be assigned to one of the following categories:

Category	Description	Examples
Hardware	Physical IT infrastructure	IBM z15, HP ProLiant DL560
IaaS	Virtualized infrastructure	AWS EC2, Azure VMs
PaaS	Cloud platforms for development	Google App Engine, SAP BTP
SaaS	Subscription-based software	Salesforce, LeanIX
Service	Managed IT services	Hosting, Support, Analytics
Software	Commercial/open-source software	.NET Framework 4.8

Versioning

Only **major versions** are modeled (e.g., `5.x`). Minor or patch versions are omitted for simplicity. If a vendor introduces significant changes in a sub-version, it may be modeled as an exception.

Component Description

The description should clearly identify the software or tool. Use the official product description from the vendor's website.

⚠ **Language: English only.**

Vendor Assignment

The vendor must be linked via a **Vendor Fact Sheet**.

⚠ Use the actual **manufacturer**, not the reseller.

Example:

- Manufacturer: TechSmith (for Camtasia)
- Reseller: CCP Software GmbH
  → Only **TechSmith** should be assigned as the vendor.

Lifecycle Management

The lifecycle should reflect:

- The vendor-defined lifecycle of the version
- The internal usage period (at least the activation date)

The responsible person must ensure lifecycle data is up to date.

⚠ Components past their end-of-life must be reviewed and risk-assessed.

Technical Capabilities

Each component must be linked to at least one **technical capability** from the LeanIX Capability Map.
Specify the **Support Type**:

- **Supports**: Component supports the capability
- **Leading**: Component is the primary implementation

If no suitable capability exists, request one via the capability process.

Technical Fit

Indicates how well the component meets business needs:

Value	Meaning
Inappropriate	Must be replaced
Unreasonable	Replacement recommended
Adequate	Optimization possible
Fully Appropriate	No action needed

This should be assessed by the technical owner, ideally in collaboration with Solution or Enterprise Architects.

IT Product Assignment

Each IT component must be linked to one or more **IT Products**.

💡 Best practice: Assign components from the IT Product perspective.

Governance & Compliance

The following must be documented:

- Export control relevance and classification
- FOSS (Free/Open Source Software) compliance

These checks must be completed **outside LeanIX** via the **PASS process**. For questions, contact the relevant compliance departments.

Initiatives

IT components can be linked to business or technical initiatives. There are currently **no modeling rules** for this linkage.

Cloud-Native Components

Cloud-native services (e.g., AWS, Azure) are **not modeled manually**. They are automatically imported and assigned via **Cloud Account Management (CAM)**.

New Versions of Existing Components

To model a new version:

- Clone the existing Fact Sheet

- Link the new version as a **successor**

Structure Summary

Field	Description	Required	Controlled Vocabulary	Notes
Name	Component name	✅	—	—
Category	Type of component	✅	Hardware, IaaS, PaaS, SaaS, Service, Software	—
Version	Major version only	✅	Format: `x.x`	Minor/patch optional only if significant
Description	English product description	✅	—	Use vendor source
Vendor	Manufacturer (not reseller)	✅	Linked Vendor Fact Sheet	—
Lifecycle	Activation & end-of-life dates	✅	—	Based on vendor lifecycle
Responsible	Person accountable for the component	✅	—	Must be assigned
Technical Capability	Linked capability	✅	From Capability Map	Support Type: Supports / Leading
Technical Fit	Suitability rating	Optional	Inappropriate, Unreasonable, Adequate, Fully Appropriate	—
Linked IT Products	Related IT products	✅	—	Assign from product side
Governance	Export control & FOSS status	✅	—	Managed via PASS
Linked Initiatives	Related initiatives	Optional	—	No modeling rules
Cloud Components	Auto-imported	—	—	Managed via CAM
Versioning	Successor linkage	✅	—	Clone & link new version

Metamodel View

The diagram shows the possible relations between the entities that can be used to model an **IT Component** in the context of the appropriate architecture level.

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LLM-Ready Structure Summary

Title: *Metamodel View – IT Component (TOGAF®-Aligned)*

Description:

This diagram represents a metamodel view of an IT system, structured according to the TOGAF® framework and suitable for modeling in LeanIX. It is divided into three horizontal layers, each corresponding to a TOGAF® architecture domain:

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1. TOGAF® A – Architecture Vision

- **Element**: `Initiative` (magenta box)
- **Relationship**: An arrow labeled **"impact on"** connects the `Initiative` to elements in the Business Architecture layer, indicating that initiatives influence business structures and requirements.

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2. TOGAF® B – Business Architecture

- **Elements**:

    - `Organisation` (magenta box)
    - `IT Product` (magenta box)
- **Relationships**:
    - `Organisation` → **"responsible / consumes"** → Technical Architecture layer
    - `IT Product` → **"uses / requires"** → Technical Architecture layer

These relationships indicate that business units (organisations) are responsible for or consume technical components, and IT products depend on them.

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3. TOGAF® C – Technical Architecture

- **Elements**:
    - `IT Component` (green box)
    - `Tech Capability` (magenta box)
- **Relationship**:
    - `Tech Capability` → **"realize"** → `IT Component`

This shows that technical capabilities are realized through specific IT components.

Additional Visual Elements:

- Three vertical circles on the right side, each aligned with one of the TOGAF® layers, visually reinforce the separation of architectural concerns.
- A legend in the bottom-right corner clarifies that arrows represent directional relationships such as responsibility, usage, realization, or impact.

Purpose: This metamodel is designed to support enterprise architecture modeling in LeanIX by clearly defining how initiatives, business structures, and technical components interrelate. It enables traceability from strategic initiatives down to the technical implementation, supporting impact analysis, capability mapping, and IT landscape planning.